

DIGITAL FORENSIC EXAMINATION REPORT

Forensic Examination of Jean Survivor's Workstation

Case Number: 001

Case Name: Data Exfiltration Case

Examiner: Haleema Abid

Date: June 22, 2026

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1. Case Overview

Case Number: 001

Case Name: Data Exfiltration Case

Examiner: Haleema Abid

Date: June 22, 2026

Description:

This case involves a data exfiltration incident at M57.Biz, a fictional company used for forensic training purposes. An unknown external attacker used social engineering techniques to deceive Jean, the company's CFO, into sending a confidential employee file via email. The file, m57biz.xls, contained sensitive personal data of 9 employees including salaries and Social Security Numbers. The attacker posed as company president Alison Smith by spoofing her email address, ultimately obtaining and leaking the data publicly. This report documents the forensic examination of a disk image acquired from Jean's workstation.

2. Evidence Examined

Image File: nps-2008-jean.E01 & nps-2008-jean.E02

Format: Expert Witness Format (E01)

Tool Used: Autopsy 4.23.0

Case Name: DataExfiltrationCase

Case Number: 001

Timezone Set: America/New_York

Ingest Modules Run:

All relevant Autopsy ingest modules were run on the image prior to analysis, including hash lookup, keyword search, email parser, recent activity, and file type identification. Results from these modules formed the basis of the investigation.

Case Details:

 Case Details

Case

Case Name:

DataExfiltrationCase

Case Number:

001

Created Date:

2026/05/08 21:24:36 (PKT)

Case Directory:

Case Type:

Single-user case

Database Name:

Case UUID:

dataexfiltrationcase 20260508 212436

Examiner

Name:

Haleema Abid

Phone:

Email:

Notes:

Practice investigation - M57.Biz data exfiltration case

Organization

Name:

Not Specified

Point of Contact:

Phone:

Email:

Edit Details

Close

3. Timeline of Events

Date & Time (PKST)	Sender	Receiver	Event
2008-07-20, 05:31:00	jean@m57.biz	alison@m57.biz	Jean sends "which email address are you using?"
2008-07-20, 05:32:51	alex@m57.biz	jean@m57.biz	Attacker (as Alex) sends "financial plans" to Jean
2008-07-20, 05:32:52 – 05:32:56	alex@m57.biz	jean@m57.biz	Attacker forwards multiple unrelated emails (CNN, Batman, UFOs, etc.)
2008-07-20, 05:33:13	alex@m57.biz	jean@m57.biz	Attacker replies "RE: which email address are you using?"
2008-07-20, 05:39:57	alison@m57.biz	jean@m57.biz	Attacker switches identity:sends "background checks" email as alison@m57.biz
2008-07-20, 05:43:48	alex <alison@m57.biz>	jean@m57.biz	Attacker sends "programmers" email: display name still "alex" but address switched to alison@m57.biz
2008-07-20, 05:43:48	alex <alison@m57.biz>	jean@m57.biz	Attacker replies "RE: which email address are you using?"

Date & Time (PKST)	Sender	Receiver	Event
2008-07-20, 05:44:00	jean@m57.biz	alex <alison@m57.biz>	Jean replies to "programmers", "background checks", and "which email address" emails
2008-07-20, 05:46:00	jean@m57.biz	alex <alison@m57.biz>	Jean replies again to multiple threads
2008-07-20, 05:50:19 – 05:50:21	alex <alison@m57.biz>	jean@m57.biz	Attacker rapidly replies to multiple threads simultaneously
2008-07-20, 07:22:45	alison@m57.biz <tuckgorge@gmail.com >	jean@m57.biz	Attacker's true identity revealed — tuckgorge@gmail.com sends "Please send me the information now" masked as alison@m57.biz
2008-07-20, 07:26:00	—	—	USB device connected to Jean's workstation
2008-07-20, 07:28:00	jean@m57.biz	alison@m57.biz <tuckgorge@gmail.com>	Jean replies "RE: Please send me the information now" — m57biz.xls attached and sent to attacker
2008-07-20, 11:03:40	alison@m57.biz <tuckgorge@gmail.com >	jean@m57.biz	Attacker sends "Thanks!" — confirming receipt of exfiltrated file
2008-07-21	AlisonM57 <alison@m57.biz>	jean@m57.biz	Real Alison raises alarm — her identity had been spoofed

Timeline Of Emails:

E-Mail From	Subject	▲ Date Received
Jean User <jean@m57.biz>	which email address are you using?	2008-07-20 05:31:00 PKST
alex <alex@m57.biz>	financial plans	2008-07-20 05:32:51 PKST
alex <alex@m57.biz>	FW: Fans ready to stay up all 'Knight' for Batman movie	2008-07-20 05:32:52 PKST
alex <alex@m57.biz>	FW: All In All, I Feel Like Another Brick In the Wall	2008-07-20 05:32:52 PKST
alex <alex@m57.biz>	FW: The CNN Political Ticker AM for Friday, July 18, 20...	2008-07-20 05:32:53 PKST
alex <alex@m57.biz>	FW: UFOs Over U.S. Military Sites?	2008-07-20 05:32:54 PKST
alex <alex@m57.biz>	FW: Making People Sick AND Poor	2008-07-20 05:32:54 PKST
alex <alex@m57.biz>	FW: Subject line: Missing girl's mom borrowed a shove..	2008-07-20 05:32:54 PKST
alex <alex@m57.biz>	FW: CNN.com Daily Top 10	2008-07-20 05:32:55 PKST
alex <alex@m57.biz>	FW: This Week in Health	2008-07-20 05:32:55 PKST
alex <alex@m57.biz>	FW: The CNN Political Ticker PM, Friday, July 18, 2008	2008-07-20 05:32:56 PKST
alex <alex@m57.biz>	FW: Obama makes first trip to Afghanistan	2008-07-20 05:32:56 PKST
alex <alex@m57.biz>	RE: which email address are you using?	2008-07-20 05:33:13 PKST
alison@m57.biz <alison@m57.biz>	background checks	2008-07-20 05:39:57 PKST
alex <alison@m57.biz>	programmers	2008-07-20 05:43:48 PKST
alex <alison@m57.biz>	RE: which email address are you using?	2008-07-20 05:43:48 PKST
Jean User <jean@m57.biz>	RE: which email address are you using?	2008-07-20 05:44:00 PKST
Jean User <jean@m57.biz>	RE: programmers	2008-07-20 05:44:00 PKST

Jean User <jean@m57.biz>	RE: background checks	2008-07-20 05:44:00 PKST
Jean User <jean@m57.biz>	RE: which email address are you using?	2008-07-20 05:46:00 PKST
Jean User <jean@m57.biz>	RE: CNN.com Daily Top 10	2008-07-20 05:46:00 PKST
alex <alison@m57.biz>	RE: which email address are you using?	2008-07-20 05:50:19 PKST
alex <alison@m57.biz>	RE: programmers	2008-07-20 05:50:20 PKST
alex <alison@m57.biz>	RE: background checks	2008-07-20 05:50:20 PKST
alex <alison@m57.biz>	RE: CNN.com Daily Top 10	2008-07-20 05:50:20 PKST
alex <alison@m57.biz>	RE: which email address are you using?	2008-07-20 05:50:21 PKST
alison@m57.biz <tuckgorge@gmail.com>	Please send me the information now	2008-07-20 07:22:45 PKST
Jean User <jean@m57.biz>	RE: Please send me the information now	2008-07-20 07:28:00 PKST
alison@m57.biz <tuckgorge@gmail.com>	Thanks!	2008-07-20 11:03:40 PKST

4. Attack Method Breakdown

The attacker executed a multi-stage social engineering attack, carefully crafting a false narrative to manipulate Jean into releasing confidential company data. The attack unfolded as follows:

Stage 1 : Establishing a Fake Internal Identity

The attacker first created a fake email identity using the address **alex@m57.biz**, presenting themselves as an **m57.biz** employee named Alex. However, Alex is not a real employee of **m57.Biz**. This identity was fabricated solely to initiate contact and build a false sense of legitimacy with Jean.

Stage 2 : Escalating to Impersonating the Company President

After initial contact was established, the attacker escalated the deception by switching to the email address **alison@m57.biz**, impersonating Alison Smith, the president of **m57.biz**. This was a deliberate escalation by posing as the company president, the attacker added authority to the request, making Jean less likely to question it.

A critical forensic finding here is the display name. The real Alison Smith always communicated using the display name **AlisonM57**. Any email arriving from **alison@m57.biz** without this display name was not sent by the real Alison but by the attacker. This distinction was confirmed through careful analysis of the email headers and display names in the recovered PST file.

President Alison's Email:

AlisonM57 <alison@m57.biz>	business plan	2008-07-07 01:25:09 PKST	Say, Jean, how is the business plan coming along?I know that today is a Sunday, but ...
AlisonM57 <alison@m57.biz>	RE: this is what I ...	2008-07-07 01:25:11 PKST	EMAIL THIS EmailJean,Please do not send me links like this. I have no way of knowin...
AlisonM57 <alison@m57.biz>	By the way...	2008-07-07 01:25:14 PKST	EMAIL THISEmailhttp://www.cnn.com/2008/LIVING/worklife/06/19/too.tattooed.to....

AlisonM57 <alison@m57.biz>	what is going on?	2008-07-21 05:41:11 PKST	What are you doing?
AlisonM57 <alison@m57.biz>	are you around t...	2008-07-21 05:47:32 PKST	Jean, Something very strange is going on. Do you know anything about it?

Spoofed Emails:

alex <alison@m57.biz>	RE: which email ...	2008-07-20 05:50:19 PKST	Yes, I got this email.-----Original Message-----From: Jean User [mailto:jean@n
alex <alison@m57.biz>	RE: programmers	2008-07-20 05:50:20 PKST	Well, make it happen.-----Original Message-----From: Jean User [mailto:jean@
alex <alison@m57.biz>	RE: background ...	2008-07-20 05:50:20 PKST	What's a "sure thing." ?-----Original Message-----From: Jean User [mailto:jean@
alex <alison@m57.biz>	RE: CNN.com D...	2008-07-20 05:50:20 PKST	CNN.com Daily Top 10Sorry; I don't know why I sent that to you. -----Origine
alex <alison@m57.biz>	RE: which email ...	2008-07-20 05:50:21 PKST	Please stop this email train.-----Original Message-----From: Jean User [mailto:j
alison@m57.biz <tuckgorge@gmail.com>	Please send me ...	2008-07-20 07:22:45 PKST	Hi, Jean.I'm sorry to bother you, but I really need that information now --- this
Jean User <jean@m57.biz>	RE: Please send ...	2008-07-20 07:28:00 PKST	I've attached the information that you have requested to this email message.-
alison@m57.biz <tuckgorge@gmail.com>	Thanks!	2008-07-20 11:03:40 PKST	Jean,Thanks for the file. I'll handle it from here. to: jean@m57.bizfrom: (alison@

Stage 3 : Fabricating a Cover Story (Venture Capital)

The attacker constructed a fake venture capital story to justify why employee records were needed. This gave the request a plausible business reason, making Jean believe that sharing the employee data was a legitimate and necessary action.

Stage 4 : Silencing Jean's Suspicions

When Jean began asking questions about the unusual emails , noticing inconsistencies , the attacker panicked. To shut down Jean's suspicions, the attacker used a **"misconfiguration"** story, claiming there was a technical email issue causing the confusion. The attacker even told Jean to **"stop this email train"** ,attempting to prevent her from investigating further. This is a classic social engineering tactic: **when a victim starts to question, redirect and reassure them quickly.**

Manipulation:

From: alex <alison@m57.biz>

To:

CC:

Subject: RE: which email address are you using?

Headers

Text

HTML

RTF

Attachments (0)

Accounts

Whoops. It looks like my email was misconfigured.

My email is alison@m57.biz, not alex. Sorry about that.

From: alex <alison@m57.biz>

To:

CC:

Subject: RE: which email address are you using?

Headers

Text

HTML

RTF

Attachments (0)

Accounts

Please stop this email train.

-----Original Message-----

From: Jean User [mailto:jean@m57.biz]

Sent: Sunday, July 20, 2008 12:47 AM

To: alison@m57.biz

Subject: RE: which email address are you using?

I'm confused.

Stage 5 : Sending the Background Check Email

A background check related email was sent from **alison@m57.biz** but critically, without the **AlisonM57** display name. This confirmed that this email too was sent by the attacker, not the real Alison.

Stage 6 : Final File Request via External Email

The final request for the employee file was made from **tuckgorge@gmail.com**, an external Gmail account, masked as **alison@m57.biz**. This was the attacker's true external identity breaking through. Jean, having been thoroughly manipulated through the previous stages, sent the file without further question.

File Request:

From: alison@m57.biz <tuckgorge@gmail.com>
To: jean@m57.biz
CC:
Subject: Please send me the information now

HeadersTextHTMLRTFAttachments (0)Accounts

Hi, Jean.

I'm sorry to bother you, but I really need that information now --- this VC guy is being very insistent. Can you please reply to this email with the information I requested --- the names, salaries, and social security numbers (SSNs) of all our current employees and intended hires?

Thanks.

Alison

File Sent:

From: Jean User <jean@m57.biz>
To: alison@m57.biz
CC:
Subject: RE: Please send me the information now

HeadersTextHTMLRTFAttachments (1)Accounts

I've attached the information that you have requested to this email message.

-----Original Message-----
From: alison@m57.biz [mailto:tuckgorge@gmail.com]
Sent: Sunday, July 20, 2008 2:23 AM
To: jean@m57.biz
Subject: Please send me the information now

Hi, Jean.

I'm sorry to bother you, but I really need that information now --- this VC guy is being very insistent. Can you please reply to this email with the information I requested --- the names, salaries, and social security numbers (SSNs) of all our current employees and intended hires?

Thanks.

5. Key Evidence

5.1. m57biz.xls (The Exfiltrated File)

The file **m57biz.xls** is the central piece of evidence in this case. It is a Microsoft Excel spreadsheet containing the sensitive personal data of 9 **m57.biz** employees, including names, salaries, and Social Security Numbers (SSNs).

File Path:

/Documents and Settings/Jean/Local Settings/Application Data/Microsoft/Outlook/outlook.pst/m57biz.xls

The file was accessed on July 20 at 07:28 and was emailed to the attacker shortly after. Autopsy flagged this file under **Bad Items** during ingest, marking it as notable content requiring examiner attention.

Bad Items			1 Results
Table	Thumbnail	Summary	
			Save Table as CSV
Source	Type	Path	
 m57biz.xls	File	/img_nps-2008-jean.E01/vol_vol2/Documents and Settings/Jean/Local Settings/Application Data/Microsoft/Outlook/outlook.pst/m57biz.xls	

Metadata	
Name:	/img_nps-2008-jean.E01/vol_vol2/Documents and Settings/Jean/Local Settings/Application Data/Microsoft/Outlook/outlook.pst/m57biz.xls
Type:	Derived
MIME Type:	application/vnd.ms-excel
Size:	291840
File Name Allocation:	Allocated
Metadata Allocation:	Allocated
Modified:	2008-07-20 07:28:03 PKST
Accessed:	0000-00-00 00:00:00
Created:	2008-07-20 07:28:03 PKST
Changed:	0000-00-00 00:00:00
MD5:	e23a4eb7f2562f53e88c9dca8b26a153
SHA-256:	34456b5f714dc9d8dd23c742d54c3f5f582ecb042bc1c4d3042b88203863779f
Hash Lookup Results:	UNKNOWN
Internal ID:	63830

5.2. Email Analysis (261 Emails Analyzed)

The Outlook PST file recovered from Jean's workstation contained 261 emails, all of which were processed and analyzed through Autopsy's email ingest module. The emails provided a

complete picture of the social engineering attack from initial contact to post-exfiltration confirmation.

Key emails identified:

- Initial contact from alex@m57.biz (fake identity)
- Escalation emails from alison@m57.biz without AlisonM57 display name (attacker)
- The "misconfiguration" and "stop this email train" email from attacker
- Final file request from tuckgorge@gmail.com
- Jean's reply containing the attached m57biz.xls
- "Thanks" reply from attacker at 11:03 on July 20 confirming receipt
- July 21 alarm email from the real Alison (AlisonM57)
- Follow-up email from Bob Blackman confirming the data was leaked publicly

From: bob@m57.biz <bob@m57.biz>

To: jean@m57.biz

CC:

Subject: Hi Jean

Headers

Text

HTML

RTF

Attachments (0)

Accounts

Hi, Jean.

This is Bob. I'm one of the programmers working on the project.

Do you know anything about my social security number being posted on the Internet? Somebody just sent me email saying that my name and SSN had been posted. I don't really know what this is about.

5.3. USB Device Artifact

Autopsy recovered 14 USB device artifacts from Jean's workstation. A USB device was recorded as connected on July 20 at 07:26 , just 2 minutes before m57biz.xls was accessed and emailed. The proximity of this timestamp to the file access event is noted. However, no conclusive evidence was found to directly link the USB connection to the exfiltration. The file was sent via email, not confirmed to have been copied to the USB device.

5.4. Recent Documents

9 recent document entries were recovered from Jean's workstation. These provided supporting evidence of file access activity around the time of the incident.

Recent Documents						
Table Thumbnail Summary						
Source Name	S	C	O	Path	Date Accessed	
 Desktop.LNK				C:\Documents and Settings\Jean\Desktop	2008-07-20 07:28:04 PKST	
 m57biz.LNK				C:\Documents and Settings\Jean\Desktop\m57biz.xls	2008-07-20 07:27:42 PKST	
 My Pictures.lnk				C:\Documents and Settings\Jean\My Documents\My ...	2008-07-06 13:50:42 PKST	
 m57biz.lnk				C:\Documents and Settings\Jean\Desktop\m57biz.xls	2008-07-20 07:28:04 PKST	
 tag-cloud.lnk				C:\Documents and Settings\Jean\Desktop\tag-cloud.j...	2008-07-12 00:00:37 PKST	
 Temp.LNK				C:\Documents and Settings\Jean\Local Settings\Temp	2008-07-20 07:27:42 PKST	
 LightBlueTop.lnk				C:\Documents and Settings\Jean\My Documents\My ...	2008-07-06 13:50:42 PKST	
 t1soft.flipflops.lnk				C:\Documents and Settings\Jean\My Documents\My ...	2008-07-06 13:54:26 PKST	
 NTUSER.DAT					2008-07-06 07:37:38 PKST	

5.5. Web Downloads

3 web download artifacts were recovered. AIM (AOL Instant Messenger) was downloaded on July 18. This is consistent with normal activity , the real Alison (AlisonM57) was also confirmed to use AIM, so this download is not considered suspicious.

5.6. Web Search History

238 web search entries were analyzed. No suspicious searches were identified that would suggest Jean was knowingly involved in the data exfiltration or had prior knowledge of the attack.

5.7. Shell Bags

42 Shell Bag artifacts were recovered. Shell bags record folder navigation activity on a Windows system. These were analyzed for evidence of unusual file browsing behavior around the time of the incident

6. Findings & Conclusion

6.1. Jean: Innocent Victim

Based on the totality of evidence examined, Jean is determined to be an innocent victim of a sophisticated social engineering attack. The email chain clearly shows that Jean was systematically deceived through fake identities, a fabricated business story, and deliberate suppression of her suspicions. Her web search history contained nothing suspicious. There is no forensic evidence to suggest Jean knowingly participated in the exfiltration.

6.2. Real Alison Smith: Also a Victim

The real Alison Smith's email identity was spoofed throughout the attack. The forensic evidence confirms that the real Alison, identifiable by her consistent use of the **AlisonM57** display name, had no involvement in the attack. Upon discovering the breach, she raised the alarm on July 21. She is a victim of identity spoofing.

6.3. Attacker: External Threat

The attacker is identified as an external threat operating primarily through the email address **tuckgorge@gmail.com**. The attacker:

- Created a fake internal identity (alex@m57.biz)
- Spoofed the company president's email (alison@m57.biz)
- Constructed a false business narrative (venture capital story)
- Used psychological manipulation to silence the victim's suspicions
- Successfully obtained and leaked confidential employee data publicly

6.4. USB Device

A USB device was connected to Jean's workstation at 07:26 on July 20, two minutes before the file access and email event. While the timing is notable, no direct forensic evidence links the USB connection to the exfiltration. The confirmed method of data transfer was email.

7. Limitations / Unresolved

USB Activity: The USB device connection at 07:26 on July 20 remains unexplained. Whether data was also copied to the USB device cannot be confirmed or ruled out based on the artifacts available.

Official Solution: The official solution exists as an encrypted PDF (M57-Jean_Solution.pdf) available through Digital Corpora. Access is restricted to verified educators. All conclusions in this report are therefore based solely on the examiner's independent forensic analysis..

Alex Identity: While it is confirmed that Alex is not a real M57.Biz employee and the alex@m57.biz address was used by the attacker, the full origin and setup of this fake account could not be traced further from the artifacts available on Jean's workstation alone.

8. Reference

Garfinkel, S., Farrell, P., Roussev, V., & Dinolt, G. (2009). Bringing Science to Digital Forensics with Standardized Forensic Corpora. *Digital Investigation*, 6, S2–S11.
<https://digitalcorpora.org/corpora/scenarios/m57-jean/>